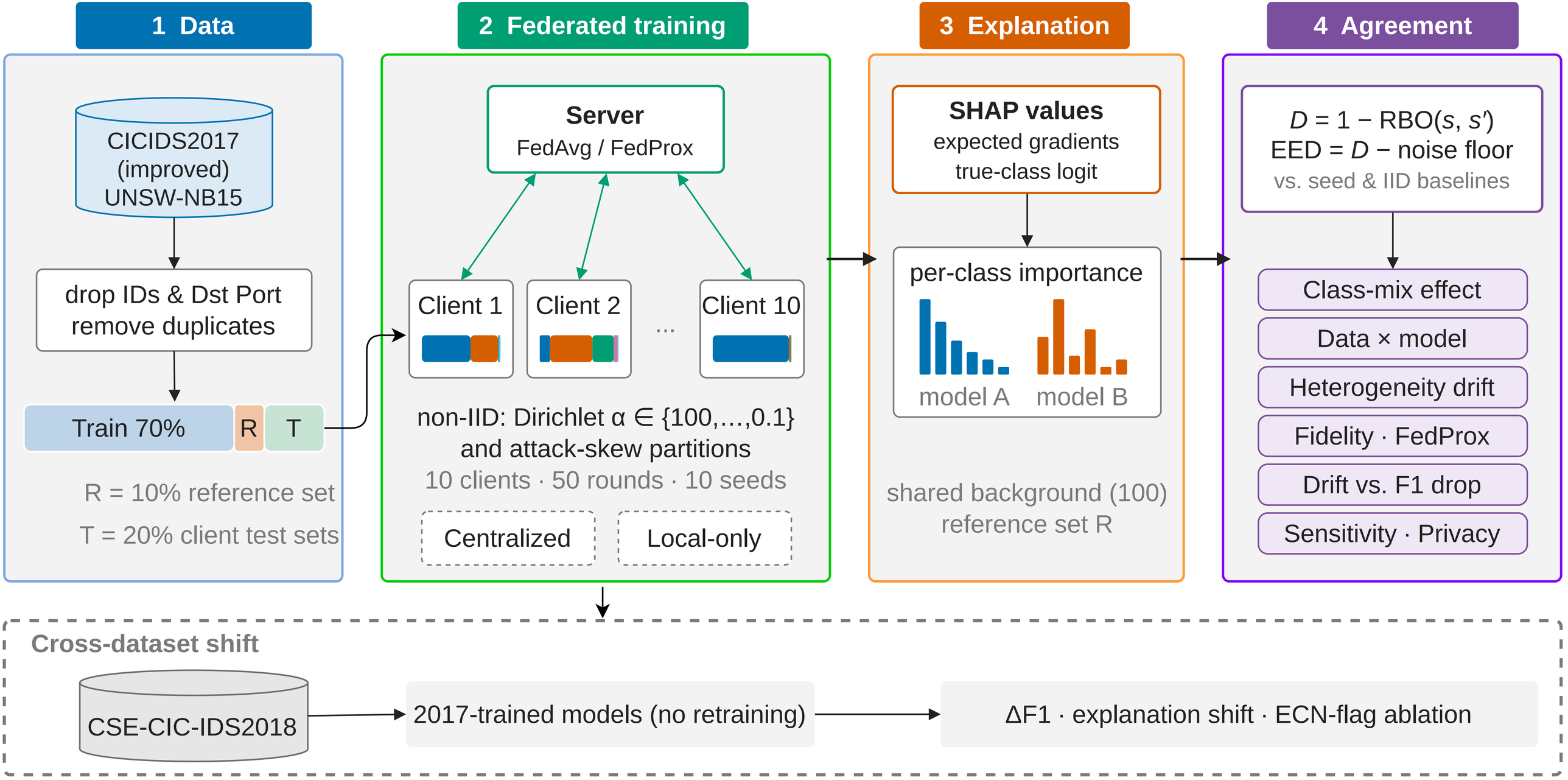

1 Data
2 Federated training
3 Explanation
4 Agreement
CICIDS2017
(improved)
UNSW-NB15
drop IDs & Dst Port
remove duplicates
Train 70%
R
T
R = 10% reference set
T = 20% client test sets
Server
FedAvg / FedProx
Client 1
Client 2
...
Client 10
non-IID: Dirichlet α ∈ {100,…,0.1}
and attack-skew partitions
10 clients · 50 rounds · 10 seeds
Centralized
Local-only
SHAP values
expected gradients
true-class logit
per-class importance
model A
model B
shared background (100)
reference set R
D = 1 − RBO(s, s′)
EED = D − noise floor
vs. seed & IID baselines
Class-mix effect
Data × model
Heterogeneity drift
Fidelity · FedProx
Drift vs. F1 drop
Sensitivity · Privacy
Cross-dataset shift
CSE-CIC-IDS2018
2017-trained models (no retraining)
ΔF1 · explanation shift · ECN-flag ablation